

ReVerie

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Figure 1: The audience went through a collective dreaming experience with their dream objects generated in real time.

ABSTRACT

ReVerie is an Interactive AI installation that collects and visualizes textual dream data from the artist and the audience and creates a collective dream-reliving experience. Within Dream science, ‘Dreamwork’ encompasses techniques such as dream analysis, interpretation, and exploration aimed at uncovering insights into the subconscious mind. Central to ‘Dreamwork’ is the concept of re-experiencing dreams—immersing oneself in the recollection of

dream memories, and emotions. Through a 3D generative diffusion model, the ‘ReVerie’ system translates the whispering of dream objects into 3D immersive visualization in real time to facilitate dream re-experiencing and a collective dream fly-through experience.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*; **Virtual reality**; • **Applied computing** → **Health care information systems**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

3D Generative AI, Dream Technology, Immersive Art Installation

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1 INTRODUCTION

Within Dream science, ‘Dreamwork’ encompasses techniques such as dream analysis, interpretation, and exploration aimed at uncovering insights into the subconscious mind. Central to ‘Dreamwork’ is the concept of re-experiencing dreams—immersing oneself in the recollection of dream memories, and emotions. Through a 3D generative diffusion model, the ‘ReVerie’ system translates the whispering of dream objects into 3D immersive visualization in real time to facilitate dream re-experiencing and a collective dream fly-through experience (Figure 1). Before the exhibition, the lead artist collab-



Figure 2: One audience member whispering her dream object into the microphone.

orated with Dream scientists and lived with the system for two weeks, while the system collected his dream objects and stored them in an anonymized dream database ‘dreamscape’.



Figure 3: (a)The ‘ReVerie’ AI system generating the dream object (b) Generated dream objects were collected in an anonymous online database and visualized in real-time.

2 3D GENERATIVE AI FOR DREAM RELIVING

The ‘ReVerie’ AI system incorporates a Generative 3D diffusion model, speech recognition, and real-time visualization.

First, a speech recognition function provided by Meta voice SDK transcribes the dream object description, which is then sent to the server. The server relays the text description as prompt input for a generative diffusion model to generate 3D PLY data. This data is then fetched from the server for the point cloud rendering through Unity Engine.

The composer composes the soundscape with a Musical Agent based on Self-Organizing Maps (MASOM) that computes valence and arousal (circumplex model of affect) through machine listening capability, based on low-level audio feature extraction from music.

3 COLLECTIVE DREAMING EXPERIENCE

During the exhibition, the audiences were invited to interact with the system through a ritualistic whispering of their dreams (Figure 2). The textual dream data and its generated point cloud data were then also stored in the ‘dreamscape’ database. Every 5 minutes after the audience interaction, the system will generate a collective dream visualization, with dream objects retrieved from the ‘dreamscape’ database (Figure 3). The audiences fly through their dream objects in the shared immersive space, along with pre-composed generative music accommodating emotional scenarios for dream re-experiencing.

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